

PF Lab 9 tasks (6)

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Task 5

```
#include<stdio.h>
```

```
void swap(int *a, int *b){
```

```
    int temp;
```

```
    temp=*a;
```

```
    *a=*b;
```

```
    *b=temp;
```

```
}
```

```
int main(){
```

```
    int a=25, b=40;
```

```
    printf("Before swap:\n");
```

```
    printf("a:%d , b:%d",a,b);
```

```
    swap(&a,&b);
```

```
    printf("\nAfter swap:\n");
```

```
    printf("a:%d , b:%d",a,b);
```

```
    return 0;
```

```
}
```

```
C:\Users\Anas Ali Khan\Documents\PF Lab9 task5.exe
Before swap:
a:25 , b:40
After swap:
a:40 , b:25
-----
Process exited after 0.01043 seconds with return value 0
Press any key to continue . . .
```

Task 6

```
#include<stdio.h>
```

```
void calculate(int a, int b, int *sum, float *avg){
```

```
    *sum=a+b;
```

```
    *avg=(a+b)/2;
```

```
}
```

```
int main()
```

```
{
```

```
    int a=20, b=30;
```

```
    int sum=0;
```

```
    float avg;
```

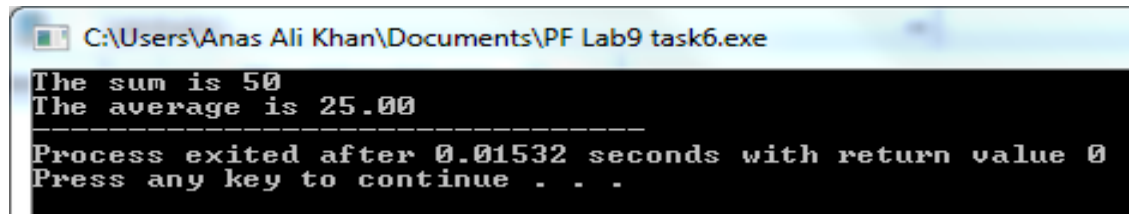
```
    calculate(a,b,&sum,&avg);
```

```
    printf("The sum is %d\n",sum);
```

```
    printf("The average is %.2f",avg);
```

```
    return 0;
```

}



```
C:\Users\Anas Ali Khan\Documents\PF Lab9 task6.exe
The sum is 50
The average is 25.00
-----
Process exited after 0.01532 seconds with return value 0
Press any key to continue . . .
```

Task 7

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    int *p=NULL;
```

```
    if(p==NULL){
```

```
        printf("The pointer doesn't reference any valid memory\n");
```

```
    }
```

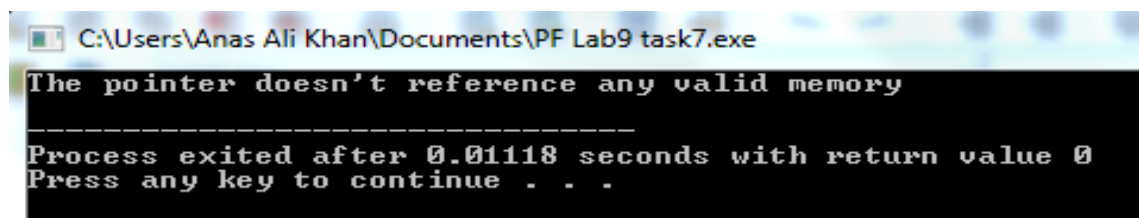
```
    else{
```

```
        printf("Pointer refers to a valid memory\n");
```

```
    }
```

```
    return 0;
```

```
}
```



```
C:\Users\Anas Ali Khan\Documents\PF Lab9 task7.exe
The pointer doesn't reference any valid memory
-----
Process exited after 0.01118 seconds with return value 0
Press any key to continue . . .
```

Task 8

```
#include<stdio.h>

int main()
{
    int a=10;

    char letter='b';

    float c=12.09;

    void *p;

    p=&a;

    printf("value of pointer is %d\n",*(int*)p);

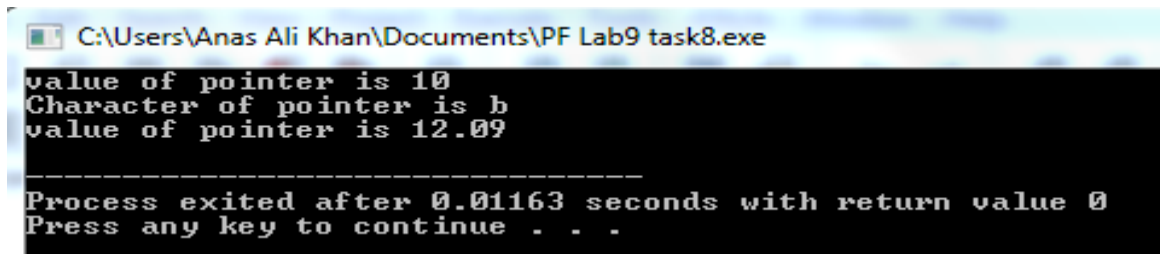
    p=&letter;

    printf("Character of pointer is %c\n",*(char*)p);

    p=&c;

    printf("value of pointer is %.2f\n",*(float*)p);

    return 0;
}
```



```
C:\Users\Anas Ali Khan\Documents\PF Lab9 task8.exe
value of pointer is 10
Character of pointer is b
value of pointer is 12.09
-----
Process exited after 0.01163 seconds with return value 0
Press any key to continue . . .
```

Task 9

```
#include <stdio.h>
```

```
float add(float a, float b) {  
    return a + b;  
}
```

```
float subtract(float a, float b) {  
    return a - b;  
}
```

```
float multiply(float a, float b) {  
    return a * b;  
}
```

```
float divide(float a, float b) {  
    if (b != 0) {  
        return a / b;  
    } else {  
        printf("Error: Division by zero!\n");  
        return 0;  
    }  
}
```

```
int main() {
```

```
int choice;

float num1, num2;

float (*operation)(float, float);

printf("Enter two numbers: ");

scanf("%f %f", &num1, &num2);


printf("1.Add 2.Subtract 3.Multiply 4.Divide\n");

printf("Choose operation (1-4): ");

scanf("%d", &choice);


switch(choice) {

    case 1: operation = add; break;

    case 2: operation = subtract; break;

    case 3: operation = multiply; break;

    case 4: operation = divide; break;

    default:

        printf("Invalid choice! Using addition.\n");

        operation = add;

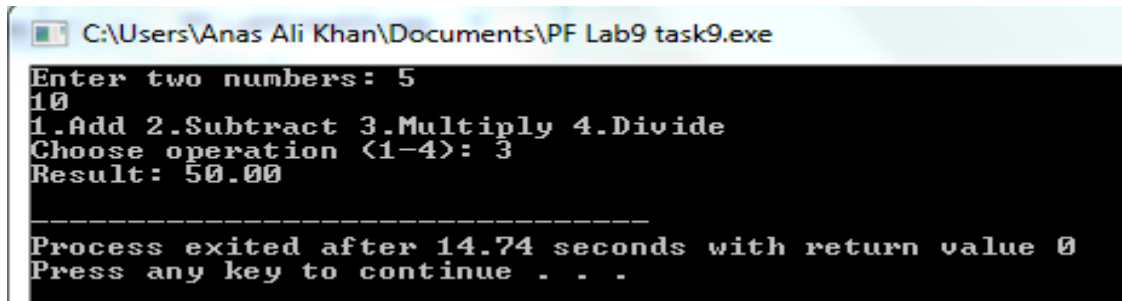
}


float result = operation(num1, num2);

printf("Result: %.2f\n", result);


return 0;
```

}



```
C:\Users\Anas Ali Khan\Documents\PF Lab9 task9.exe
Enter two numbers: 5
10
1.Add 2.Subtract 3.Multiply 4.Divide
Choose operation <1-4>: 3
Result: 50.00

-----
Process exited after 14.74 seconds with return value 0
Press any key to continue . . .
```

Task 10

```
#include<stdio.h>
```

```
float average(int a, int b, int c){
```

```
    float avg=(a+b+c)/3;
```

```
    return avg;
```

```
}
```

```
int main()
```

```
{
```

```
    int a,b,c;
```

```
    printf("Enter three numbers\n");
```

```
    scanf("%d%d%d",&a,&b,&c);
```

```
    printf("The average of these three values is %.2f",average(a,b,c));
```

```
    return 0;
```

```
}
```


C:\Users\Anas Ali Khan\Documents\PF Lab9 task10.exe

Enter three numbers

2

4

6

The average of these three values is 4.00

Process exited after 8.56 seconds with return value 0

Press any key to continue . . .